Tugas 4

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FAKULTAS INFORMATIKA & PARIWISATA

PRODI TEKNOLOGI INFORMASI

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Latihan Pembelajaran

Pada latihan ini, akan membangun sistem sederhana berbasis Event-Driven Architecture, di mana:

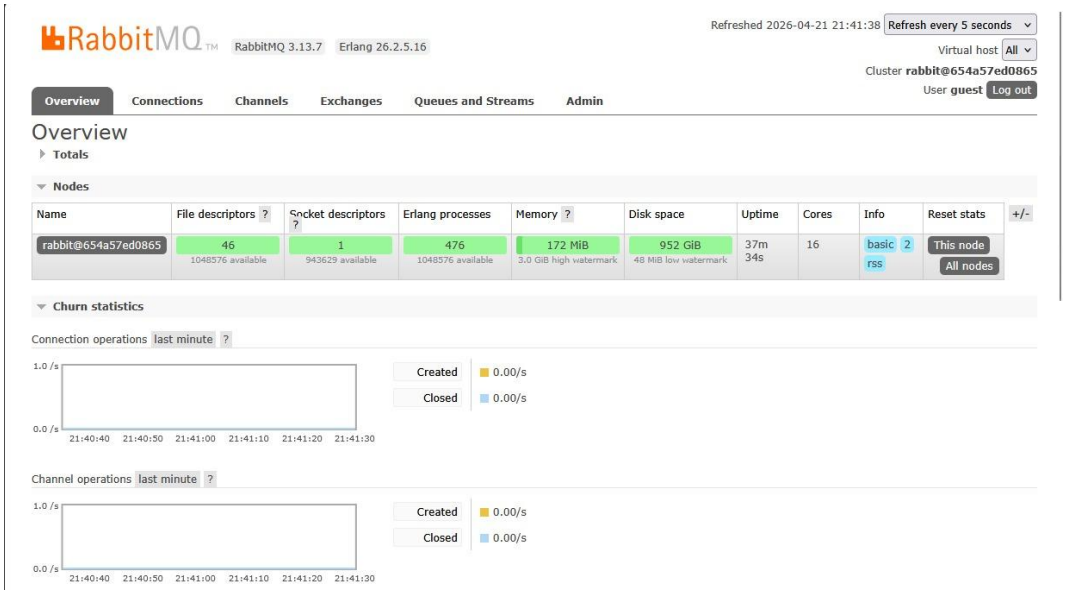
- Book Service bertindak sebagai Producer (Publisher)
- User Service bertindak sebagai Consumer (Subscriber)

Ketika data buku ditambahkan, sistem akan mengirim event BookAdded yang kemudian diproses oleh service lain.

Instalasi RabbitMQ

```
C:\Users\Maulana Fatullah>docker run -d --name rabbitmq -p 5672:5672 -p 15672:15672 rabbitmq:3-management
Unable to find image 'rabbitmq:3-management' locally
3-management: Pulling from library/rabbitmq
d686b50ce172: Pull complete
20043066d3d5: Pull complete
efa05c660812: Pull complete
7084ac8f4ab6: Pull complete
d63c4408fdb7: Pull complete
beb0d870d438: Pull complete
f4e7f62f4b80: Pull complete
5e1c03b67a78: Pull complete
14e6209d5fa8: Pull complete
3fadc669ef11: Pull complete
7ec4c696b8bb: Download complete
16ef6b12768c: Download complete
Digest: sha256:e582c0bc7766f3342496d8485efb5a1df782b5ce3886ad017e2eaae442311f69
Status: Downloaded newer image for rabbitmq:3-management
654a57ed0865bef286c492feb4d5e4ba8efed87c8a07a0920d28d94664c0bd98
```

Menjalankan RabbitMQ menggunakan Docker



Mengakses Dashboard RabbitMQ dengan URL http://localhost:15672/

```
$ npm install amqplib

added 2 packages, and audited 190 packages in 2s

found 0 vulnerabilities
```

Install Dependency pada setiap Microservices

```
book-services > .env
You, 34 minutes ago | 1 author (You)
1  API_PORT=3001
2  NODE_ENV=production
3  EXPRESS_DB_HOST=localhost
4  EXPRESS_DB_USERNAME=root
5  EXPRESS_DB_PASSWORD=
6  EXPRESS_DB_NAME=pw_praktikum_db
7  EXPRESS_DB_PORT=3306
8  RABBITMQ_URL=amqp://localhost:5672
```

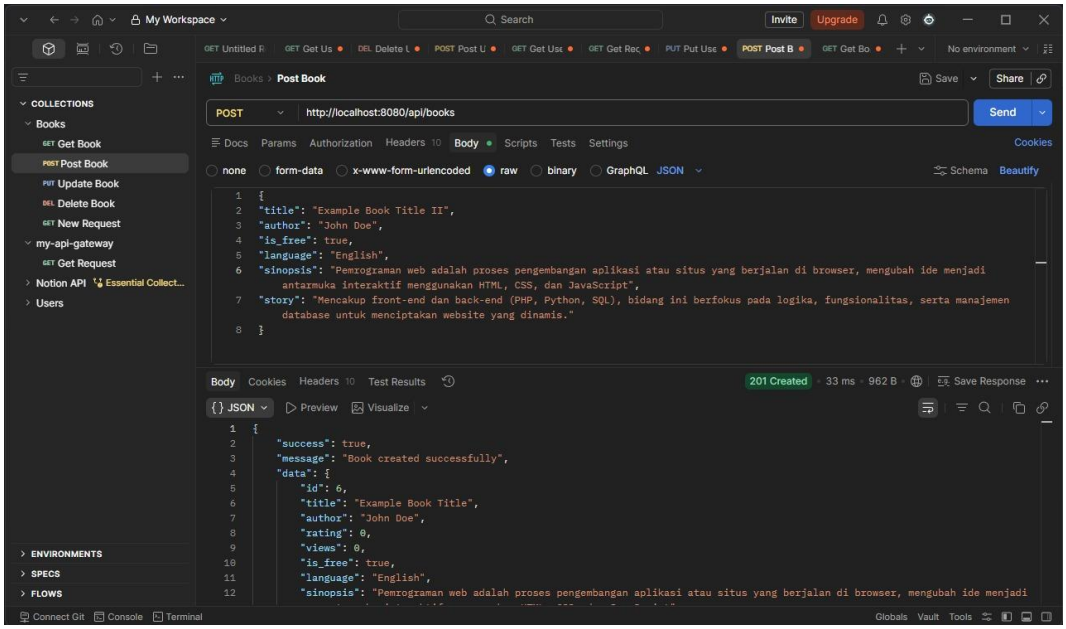
Menambahkan konfigurasi RabbitMQ di file .env

```
const amqp = require('amqplib');
class RabbitMQService {
    constructor() {
        this.connection = null;
        this.channel = null;
    }
    async connect() {
        try {
            this.connection = await amqp.connect(process.env.RABBITMQ_URL ||
            this.channel = await this.connection.createChannel();
            console.log('Connected to RabbitMQ');
            return true;
        } catch (error) {
            console.error('Failed to connect to RabbitMQ:', error);
            return false;

    }
    async sendToQueue(queueName, message) {
        try {
            if (!this.channel) {
                await this.connect();
            }
            await this.channel.assertQueue(queueName, {
                durable: true
            });
            this.channel.sendToQueue(
                queueName,
                Buffer.from(JSON.stringify(message)),
                {
                    persistent: true
                }
            );
            console.log(`Message sent to queue ${queueName}:`, message);
            return true;
        } catch (error) {
            console.error('Error sending message to queue:', error);
            return false;
        }
    }
    async close() {
        try {
            if (this.channel) await this.channel.close();
            if (this.connection) await this.connection.close();
            console.log('RabbitMQ connection closed');
        } catch (error) {
            console.error('Error closing RabbitMQ connection:', error);
        }
    }
}
module.exports = new RabbitMQService();
```

Pasang konfigurasi RabbitMQ pada config/rabbitmq.js

```jsx
};

exports.createBook = async (req, res) => {
    try {
        const {
            title,
            author,
            rating,
            views,
            is_free,
            language,
            sinopsis,
            story,
            image,
        } = req.body;
        // Validation
        if (!title || !author || !sinopsis || !story) {
            return res.status(400).json({
                success: false,
                message: "Title, author, sinopsis, and story are required",
            });
        }
        const book = await Book.create({
            title,
            author,
            rating: rating || 0,
            views: views || 0,
            is_free: is_free || false,
            language: language || "English",
            sinopsis,
            story,
            image,
        });
        // send data to message broker
        await rabbitmq.sendToQueue("book_created", {
            event: "book.created",
            timestamp: new Date().toISOString(),
            data: {
                id: book.id,
                title: book.title,
                author: book.author,
                is_free: book.is_free,
                language: book.language,
            },
        });
        //end of message broker
        res.status(201).json({
            success: true,
            message: "Book created successfully",
            data: book,
        });
    } catch (error) {
        console.error("Error creating book:", error);
        // Handle validation errors
        if (error.name === "SequelizeValidationError") {
            return res.status(400).json({
                success: false,
                message: "Validation error",
                errors: error.errors.map((e) => ({
                    field: e.path,
                    message: e.message,
                })),
            });
        }
        res.status(500).json({
            success: false,
            message: "Failed to create book",
            error: error.message,
        });
    }
};
```

Implementasi Book Services



Mengirim Request ke Book Service

```
Executing (default): INSERT INTO `books`
Executing (default): INSERT INTO `books`
 VALUES (DEFAULT,?,?,?,?,?,?,?,?,?,?);
Message sent to queue book_created: {
  event: 'book.created',
  timestamp: '2026-04-21T14:25:47.897Z',
  data: {
    id: 6,
    title: 'Example Book Title',
    author: 'John Doe',
    is_free: true,
    language: 'English'
  }
}
```

Respon Message Broker (producer)

Command Prompt

```
2026-04-21 14:04:25.268160+00:00 [info] <0.254.0> Running boot step rabbit_policies defined by app rabbit
2026-04-21 14:04:25.268624+00:00 [info] <0.254.0> Running boot step rabbit_policy defined by app rabbit
2026-04-21 14:04:25.268767+00:00 [info] <0.254.0> Running boot step rabbit_queue_location_validator defined by app rabbit
2026-04-21 14:04:25.269040+00:00 [info] <0.254.0> Running boot step rabbit_quorum_memory_manager defined by app rabbit
2026-04-21 14:04:25.269130+00:00 [info] <0.254.0> Running boot step rabbit_quorum_queue defined by app rabbit
2026-04-21 14:04:25.269546+00:00 [info] <0.254.0> Running boot step rabbit_stream_coordinator defined by app rabbit
2026-04-21 14:04:25.269996+00:00 [info] <0.254.0> Running boot step rabbit_vhost_limit defined by app rabbit
2026-04-21 14:04:25.270294+00:00 [info] <0.254.0> Running boot step rabbit_federation_parameters defined by app rabbitmq_federation
2026-04-21 14:04:25.270755+00:00 [info] <0.254.0> Running boot step rabbit_federation_supervisor defined by app rabbitmq_federation
2026-04-21 14:04:25.289205+00:00 [info] <0.254.0> Running boot step rabbit_federation_queue defined by app rabbitmq_federation
2026-04-21 14:04:25.289519+00:00 [info] <0.254.0> Running boot step rabbit_federation_upstream_exchange defined by app rabbitmq_federation
2026-04-21 14:04:25.289717+00:00 [info] <0.254.0> Running boot step rabbit_mgmt_reset_handler defined by app rabbitmq_management
2026-04-21 14:04:25.289878+00:00 [info] <0.254.0> Running boot step rabbit_mgmt_db_handler defined by app rabbitmq_management_agent
2026-04-21 14:04:25.289983+00:00 [info] <0.254.0> Management plugin: using rates mode 'basic'
2026-04-21 14:04:25.290620+00:00 [info] <0.254.0> Running boot step recovery defined by app rabbit
2026-04-21 14:04:25.320709+00:00 [info] <0.254.0> Running boot step empty_db_check defined by app rabbit
2026-04-21 14:04:25.320849+00:00 [info] <0.254.0> Will seed default virtual host and user...
2026-04-21 14:04:25.321092+00:00 [info] <0.254.0> Adding vhost '/' (description: 'Default virtual host', tags: [])
2026-04-21 14:04:25.356712+00:00 [info] <0.642.0> Making sure data directory '/var/lib/rabbitmq/mnesia/rabbit@654a57ed0065/msg_stores/vhosts/628WB79CIFDYO9LJI6DKMI09L' for vhost '/' exists
2026-04-21 14:04:25.361349+00:00 [info] <0.642.0> Setting segment_entry_count for vhost '/' with 0 queues to '2048'
2026-04-21 14:04:25.388673+00:00 [info] <0.642.0> Starting message stores for vhost '/'
2026-04-21 14:04:25.388953+00:00 [info] <0.651.0> Message store "628WB79CIFDYO9LJI6DKMI09L/msg_store_transient": using rabbit_msg_store_ets_index to provide index
2026-04-21 14:04:25.392859+00:00 [info] <0.642.0> Started message store of type transient for vhost '/'
2026-04-21 14:04:25.393050+00:00 [info] <0.655.0> Message store "628WB79CIFDYO9LJI6DKMI09L/msg_store_persistent": using rabbit_msg_store_ets_index to provide index
2026-04-21 14:04:25.394823+00:00 [warning] <0.655.0> Message store "628WB79CIFDYO9LJI6DKMI09L/msg_store_persistent": rebuilding indices from scratch
2026-04-21 14:04:25.396780+00:00 [info] <0.642.0> Started message store of type persistent for vhost '/'
2026-04-21 14:04:25.397068+00:00 [info] <0.642.0> Recovering 0 queues of type rabbit_classic_queue took 34ms
2026-04-21 14:04:25.397134+00:00 [info] <0.642.0> Recovering 0 queues of type rabbit_quorum_queue took 0ms
2026-04-21 14:04:25.397269+00:00 [info] <0.642.0> Recovering 0 queues of type rabbit_stream_queue took 0ms
2026-04-21 14:04:25.403701+00:00 [info] <0.254.0> Created user 'guest'
2026-04-21 14:04:25.407295+00:00 [info] <0.254.0> Successfully set user tags for user 'guest' to [administrator]
2026-04-21 14:04:25.411596+00:00 [info] <0.254.0> Successfully set permissions for user 'guest' in virtual host '/' to '.*', '.*', '.*'
2026-04-21 14:04:25.411734+00:00 [info] <0.254.0> Running boot step rabbit_observer_cli defined by app rabbit
2026-04-21 14:04:25.411959+00:00 [info] <0.254.0> Running boot step rabbit_looking_glass defined by app rabbit
2026-04-21 14:04:25.412160+00:00 [info] <0.254.0> Running boot step rabbit_core_metrics_gc defined by app rabbit
2026-04-21 14:04:25.412516+00:00 [info] <0.254.0> Running boot step background_gc defined by app rabbit
2026-04-21 14:04:25.412887+00:00 [info] <0.254.0> Running boot step routing_ready defined by app rabbit
2026-04-21 14:04:25.412997+00:00 [info] <0.254.0> Running boot step pre_flight defined by app rabbit
2026-04-21 14:04:25.413135+00:00 [info] <0.254.0> Running boot step notify_cluster defined by app rabbit
2026-04-21 14:04:25.413335+00:00 [info] <0.254.0> Running boot step networking defined by app rabbit
2026-04-21 14:04:25.413474+00:00 [info] <0.254.0> Running boot step rabbit_quorum_queue_periodic_membership_reconciliation defined by app rabbit
2026-04-21 14:04:25.413725+00:00 [info] <0.254.0> Running boot step definition_import_worker_pool defined by app rabbit
2026-04-21 14:04:25.413898+00:00 [info] <0.315.0> Starting worker pool 'definition_import_pool' with 16 processes in it
2026-04-21 14:04:25.414997+00:00 [info] <0.254.0> Running boot step cluster_name defined by app rabbit
2026-04-21 14:04:25.415195+00:00 [info] <0.254.0> Initialising internal cluster ID to 'rabbitmq-cluster-id-CHWetci_PHH1lphHABW7Q'
2026-04-21 14:04:25.419086+00:00 [info] <0.254.0> Running boot step virtual_host_reconciliation defined by app rabbit
2026-04-21 14:04:25.419252+00:00 [info] <0.254.0> Running boot step direct_client defined by app rabbit
2026-04-21 14:04:25.419368+00:00 [info] <0.254.0> Running boot step rabbit_federation_exchange defined by app rabbitmq_federation
2026-04-21 14:04:25.419672+00:00 [info] <0.254.0> Running boot step rabbit_management_load_definitions defined by app rabbitmq_management
2026-04-21 14:04:25.419888+00:00 [info] <0.704.0> Resetting node maintenance status
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0> Deprecated features: `management_metrics_collection`: Feature `management_metrics_collection` is deprecated.
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0> By default, this feature can still be used for now.
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0> Its use will not be permitted by default in a future minor RabbitMQ version and the feature will be removed from a future major RabbitMQ ve
rsion; actual versions to be determined.
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0> To continue using this feature when it is not permitted by default, set the following parameter in your configuration:
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0>     "deprecated_features.permit.management_metrics_collection = true"
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0> To test RabbitMQ as if the feature was removed, set this in your configuration:
2026-04-21 14:04:25.810121+00:00 [warning] <0.733.0>     "deprecated_features.permit.management_metrics_collection = false"
2026-04-21 14:04:32.002639+00:00 [info] <0.770.0> Management plugin: HTTP (non-TLS) listener started on port 15672
2026-04-21 14:04:32.012879+00:00 [info] <0.800.0> Statistics database started.
2026-04-21 14:04:32.002994+00:00 [info] <0.799.0> Starting worker pool 'management_worker_pool' with 3 processes in it
2026-04-21 14:04:32.016693+00:00 [info] <0.818.0> Prometheus metrics: HTTP (non-TLS) listener started on port 15692
2026-04-21 14:04:32.016944+00:00 [info] <0.704.0> Ready to start client connection listeners
2026-04-21 14:04:32.020059+00:00 [info] <0.862.0> started TCP listener on [::]:5672
 completed with 5 plugins.
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0> Server startup complete; 5 plugins started.
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0>  * rabbitmq_prometheus
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0>  * rabbitmq_federation
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0>  * rabbitmq_management
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0>  * rabbitmq_management_agent
2026-04-21 14:04:32.256145+00:00 [info] <0.704.0>  * rabbitmq_web_dispatch
2026-04-21 14:04:32.323507+00:00 [info] <0.9.0> Time to start RabbitMQ: 17702 ms
2026-04-21 14:05:04.920663+00:00 [info] <0.894.0> accepting AMQP connection <0.894.0> (172.17.0.1:54894 -> 172.17.0.2:5672)
2026-04-21 14:05:04.921051+00:00 [error] <0.894.0> closing AMQP connection <0.894.0> (172.17.0.1:54894 -> 172.17.0.2:5672):
2026-04-21 14:05:04.921051+00:00 [error] <0.894.0> {bad_header,<<"GET / HT">>}
2026-04-21 14:05:04.984527+00:00 [info] <0.900.0> accepting AMQP connection <0.900.0> (172.17.0.1:54910 -> 172.17.0.2:5672)
2026-04-21 14:05:04.984794+00:00 [error] <0.900.0> closing AMQP connection <0.900.0> (172.17.0.1:54910 -> 172.17.0.2:5672):
2026-04-21 14:05:04.984794+00:00 [error] <0.900.0> {bad_header,<<"GET /fav">>}
2026-04-21 14:05:12.117852+00:00 [warning] <0.904.0> HTTP access denied: user 'guest' - invalid credentials
2026-04-21 14:05:14.413045+00:00 [info] <0.929.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:14.413213+00:00 [info] <0.929.0> Successfully synced tables from a peer
2026-04-21 14:05:23.337784+00:00 [info] <0.951.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:23.337995+00:00 [info] <0.951.0> Successfully synced tables from a peer
2026-04-21 14:05:25.166529+00:00 [info] <0.963.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:25.166750+00:00 [info] <0.963.0> Successfully synced tables from a peer
2026-04-21 14:05:34.210385+00:00 [warning] <0.973.0> Resetting RabbitMQ management storage
2026-04-21 14:05:34.240202+00:00 [info] <0.984.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:34.240442+00:00 [info] <0.984.0> Successfully synced tables from a peer
2026-04-21 14:05:37.230403+00:00 [info] <0.991.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:37.230637+00:00 [info] <0.991.0> Successfully synced tables from a peer
2026-04-21 14:05:39.787626+00:00 [info] <0.1003.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:05:39.787988+00:00 [info] <0.1003.0> Successfully synced tables from a peer
2026-04-21 14:07:05.421782+00:00 [info] <0.1152.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:07:05.422040+00:00 [info] <0.1152.0> Successfully synced tables from a peer
2026-04-21 14:22:38.715022+00:00 [info] <0.2107.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:22:38.715647+00:00 [info] <0.2107.0> Successfully synced tables from a peer
2026-04-21 14:23:45.387169+00:00 [info] <0.2174.0> accepting AMQP connection <0.2174.0> (172.17.0.1:60182 -> 172.17.0.2:5672)
2026-04-21 14:23:45.448304+00:00 [info] <0.2174.0> connection <0.2174.0> (172.17.0.1:60182 -> 172.17.0.2:5672): user 'guest' authenticated and granted access to vhost '/'
2026-04-21 14:24:17.732411+00:00 [info] <0.2249.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:17.732653+00:00 [info] <0.2249.0> Successfully synced tables from a peer
2026-04-21 14:24:23.296543+00:00 [info] <0.2269.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:23.296751+00:00 [info] <0.2269.0> Successfully synced tables from a peer
2026-04-21 14:24:25.507427+00:00 [info] <0.2279.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:25.507606+00:00 [info] <0.2279.0> Successfully synced tables from a peer
2026-04-21 14:24:27.515188+00:00 [info] <0.2287.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:27.515368+00:00 [info] <0.2287.0> Successfully synced tables from a peer
2026-04-21 14:24:28.432823+00:00 [info] <0.2301.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:28.433112+00:00 [info] <0.2301.0> Successfully synced tables from a peer
2026-04-21 14:24:38.552668+00:00 [info] <0.2319.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:24:38.553054+00:00 [info] <0.2319.0> Successfully synced tables from a peer
2026-04-21 14:25:39.255405+00:00 [info] <0.2402.0> Waiting for Mnesia tables for 30000 ms, 9 retries left
2026-04-21 14:25:39.255715+00:00 [info] <0.2402.0> Successfully synced tables from a peer

C:\Users\Maulana Fatullah>rabbitmq
'rabbitmq' is not recognized as an internal or external command,
operable program or batch file.

C:\Users\Maulana Fatullah>docker rabbitmq
docker: unknown command: docker rabbitmq

Run 'docker --help' for more information

C:\Users\Maulana Fatullah>docker exec -it rabbitmq rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
book_created    3

C:\Users\Maulana Fatullah>_
```

```
C:\Users\Maulana Fatullah>docker exec -it rabbitmq rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
book_created    3

C:\Users\Maulana Fatullah>docker exec -it rabbitmq rabbitmqctl list_queues
Timeout: 60.0 seconds ...
Listing queues for vhost / ...
name    messages
book_created    2
```

Mengecek pada Message Broker